

This document describes the requirements for the research project deliverables, namely the one-page research project proposal and the final report.

Due dates. If you are on the research track of 598, you *must* submit a proposal meeting the requirements listed below by March 27. Then, the final research proposal submission will be due April 27.

1 Proposal Requirements

The main purpose of the proposal is to give you an opportunity to get feedback on the idea and make sure you are on track to reach a meaningful milestone by the end of the semester. Your proposal must be **at most one page** and should describe your project idea in enough detail that it can be meaningfully evaluated by your peers and the course staff. The proposal should make a clear connection between your project and the content covered in the course, and it should articulate the research novelty of your idea or explain why the task you are undertaking is important. A key part of the proposal is **differentiating your idea from prior work**. You should discuss related work and explain how your project builds on or extends what has been done before.

Your proposal should also include a **plan for completing the project**, with expected milestones and a discussion of risk mitigation: if something doesn't work as expected, what will you do instead? If you envision this project as the first step of a larger effort, explain what you will accomplish by April 27 and outline the steps you plan to take afterwards. You should clearly state your **expected deliverables**: will you produce a write-up, working code, benchmarks, a formal proof, or some combination? You should also describe how you plan to **evaluate** your project—for implementation-oriented projects, this might include metrics like proof size, prover time, or constraint count; for more theoretical projects, explain what would constitute a meaningful result.

Additionally, explain what applications your idea might have, if any, and provide some evidence of **feasibility**: for example, preliminary exploration you have done, a toy example you have tried, or a relevant library you have identified.

Finally, if you need any resources such as server time or specialized hardware, or anticipate needing help from the course staff, explain what you need and why.

Team projects. Based on prior discussions with students, the expectation of the course staff is that most projects will be completed by a single student. However, if you want to work in a team, the course staff will consider this on a case-by-case basis. Include in your proposal information about the proposed team makeup and division of labor.

2 Final Report Requirements

Your final report should be a self-contained document written in the style of a research paper. It should be in single-column format with one-inch margins and 11pt Times font. The report should follow standard paper structure: introduction, related work, technical approach, evaluation or results, and conclusion. The **target length is 8–10 pages** excluding references; if you believe your project requires more space, contact the course staff for permission.

Artifacts. Along with your report, you must submit an artifact that can be used to reproduce your results. This could be a code repository, a set of scripts, a proof artifact, or any other material needed to verify your claims. Include clear instructions for building and running your artifact.

Evaluation. If your project is implementation-oriented, the evaluation section is especially important. A strong evaluation section for an applied cryptographic engineering or systems paper should begin by clearly stating the **goals** of the evaluation: what are the key questions you are trying to answer? For example, you might ask how proof generation time scales with circuit size, how your approach compares to a baseline, or whether a particular optimization yields meaningful improvement. After stating the goals, describe your **methodology**: what experiments did you run, on what hardware, with what parameters, and how did you control for variance? Finally, present your results and interpret them in the context of the questions you posed. Figures and tables summarizing performance data are strongly encouraged. Even for more theoretical projects, you should present your results clearly and explain their significance.

Reflection and postmortem. Your report should include a candid reflection on the project. What worked well? What did not pan out, and why? If you changed direction during the project, explain what motivated the change. This section should also include suggestions for future work: if someone wanted to continue this project, what would be the most promising next steps or alternative approaches?

Comparison to proposal. Briefly discuss how the final project compares to what you originally proposed. If the scope changed, explain why and what was actually accomplished relative to the original plan.

Longer-term projects. If your research project is part of a longer-term effort, describe your next steps and your eventual goal for the work (e.g., submission to a specific conference or workshop).

Presentations. Presentations are not required, but if you would like to present your work to the class, contact the course staff to arrange a slot.